

5. Mortality

Contents

1	Environment preparation	1
2	Counts and cumulative proportions	1
2.1	By lineage, tumoral status, and sleep condition	1
2.2	By lineage and tumoral status	2
2.3	By lineage and sleep condition	2
2.4	By tumoral status and sleep condition	3
2.5	By lineage	3
2.6	By tumoral status	4
2.7	By sleep condition	4

1 Environment preparation

Run scripts with the needed packages and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
```

2 Counts and cumulative proportions

2.1 By lineage, tumoral status, and sleep condition

```
df %>%
  group_by(lineage, tum_state, sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 18 x 6
##   lineage tum_state sleep_condition n_total n_dead prop
##   <fct>   <fct>     <fct>         <int>  <dbl> <dbl>
```

##	1	HC_MT	Healthy	Control	48	1	2.08
##	2	HC_MT	Healthy	Mechanical sleep disruption	48	6	12.5
##	3	HO_MT	Healthy	Control	62	1	1.61
##	4	HO_MT	Healthy	Mechanical sleep disruption	38	1	2.63
##	5	HO_MT	Tumoral	Control	10	0	0
##	6	HO_MT	Tumoral	Mechanical sleep disruption	34	1	2.94
##	7	HO_SPC	Healthy	Control	62	2	3.23
##	8	HO_SPC	Healthy	Mechanical sleep disruption	29	1	3.45
##	9	HO_SPC	Tumoral	Control	10	0	0
##	10	HO_SPC	Tumoral	Mechanical sleep disruption	43	0	0
##	11	HO_SPT	Healthy	Control	12	5	41.7
##	12	HO_SPT	Healthy	Mechanical sleep disruption	25	3	12
##	13	HO_SPT	Tumoral	Control	60	2	3.33
##	14	HO_SPT	Tumoral	Mechanical sleep disruption	47	0	0
##	15	HO_VLN	Healthy	Control	96	1	1.04
##	16	HO_VLN	Healthy	Mechanical sleep disruption	96	0	0
##	17	HV_GAL	Healthy	Control	72	1	1.39
##	18	HV_GAL	Healthy	Mechanical sleep disruption	72	6	8.33

2.2 By lineage and tumoral status

```
df %>%
  group_by(lineage, tum_state) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 9 x 5
##   lineage tum_state n_total n_dead prop
##   <fct>   <fct>      <int>  <dbl> <dbl>
## 1 HC_MT   Healthy        96      7  7.29
## 2 HO_MT   Healthy       100      2   2
## 3 HO_MT   Tumoral        44      1  2.27
## 4 HO_SPC  Healthy        91      3  3.3
## 5 HO_SPC  Tumoral        53      0   0
## 6 HO_SPT  Healthy        37      8 21.6
## 7 HO_SPT  Tumoral       107      2  1.87
## 8 HO_VLN  Healthy       192      1  0.52
## 9 HV_GAL  Healthy       144      7  4.86
```

2.3 By lineage and sleep condition

```
df %>%
  group_by(lineage, sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
```

```
prop = round(n_dead / n_total * 100, 2),
.groups = "drop")
```

```
## # A tibble: 12 x 5
##   lineage sleep_condition      n_total n_dead prop
##   <fct>    <fct>          <int>   <dbl> <dbl>
## 1 HC_MT    Control             48        1  2.08
## 2 HC_MT    Mechanical sleep disruption  48        6 12.5
## 3 HO_MT    Control             72        1  1.39
## 4 HO_MT    Mechanical sleep disruption  72        2  2.78
## 5 HO_SPC   Control             72        2  2.78
## 6 HO_SPC   Mechanical sleep disruption  72        1  1.39
## 7 HO_SPT   Control             72        7  9.72
## 8 HO_SPT   Mechanical sleep disruption  72        3  4.17
## 9 HO_VLN   Control             96        1  1.04
## 10 HO_VLN  Mechanical sleep disruption  96        0  0
## 11 HV_GAL   Control             72        1  1.39
## 12 HV_GAL  Mechanical sleep disruption  72        6  8.33
```

2.4 By tumoral status and sleep condition

```
df %>%
  group_by(tum_state, sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 4 x 5
##   tum_state sleep_condition      n_total n_dead prop
##   <fct>    <fct>          <int>   <dbl> <dbl>
## 1 Healthy   Control             352       11  3.12
## 2 Healthy   Mechanical sleep disruption  308       17  5.52
## 3 Tumoral   Control              80        2   2.5
## 4 Tumoral   Mechanical sleep disruption  124        1  0.81
```

2.5 By lineage

```
df %>%
  group_by(lineage) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 6 x 4
```

```
## lineage n_total n_dead prop
## <fct> <int> <dbl> <dbl>
## 1 HC_MT 96 7 7.29
## 2 HO_MT 144 3 2.08
## 3 HO_SPC 144 3 2.08
## 4 HO_SPT 144 10 6.94
## 5 HO_VLN 192 1 0.52
## 6 HV_GAL 144 7 4.86
```

2.6 By tumoral status

```
df %>%
  group_by(tum_state) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 2 x 4
## tum_state n_total n_dead prop
## <fct> <int> <dbl> <dbl>
## 1 Healthy 660 28 4.24
## 2 Tumoral 204 3 1.47
```

2.7 By sleep condition

```
df %>%
  group_by(sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 2 x 4
## sleep_condition n_total n_dead prop
## <fct> <int> <dbl> <dbl>
## 1 Control 432 13 3.01
## 2 Mechanical sleep disruption 432 18 4.17
```